

Experiment 7: Measurement of Young's Modulus using Strain Gauges

Laboratory Equipment

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| 1) Cantilevered beam with mounted strain gauges. | 6) Screw Driver, Pliers, Wire Stripper |
| 2) AD622 Instrumentation Amp | 7) Precision Matched Resistors |
| 3) Powered Breadboard & ± 15 Volt source | 8) Calipers |
| 4) Table-top Digital Multi-Meter | 9) Measuring Tape |
| 5) Wire & Resistor Kits | 10) Weight Set |

Background

Measurement of strain is a fundamental tool used by engineers in both research and design. Strain measurements are often used in systems where force or pressure measurements are required. In some applications the strain (and associated stress) of a part can only be determined by measurement, as calculation is either too complicated or not accurate enough. In this experiment, you will learn to use a metallic strain gauge to measure strain in an end-loaded cantilevered beam. Simultaneously, you will calculate the stress due to bending caused by the end load. You will use these values to create a stress v. strain plot to determine the modulus of elasticity (or Young's modulus) of the beam.

Laboratory Objectives

- 1) Assemble a Wheatstone bridge circuit to measure a signal from two strain gauges affixed to a cantilevered beam.
- 2) Collect enough strain and loading data to perform a linear regression to estimate Young's modulus, E , of the cantilevered beam. Compare the experimental value with published values.

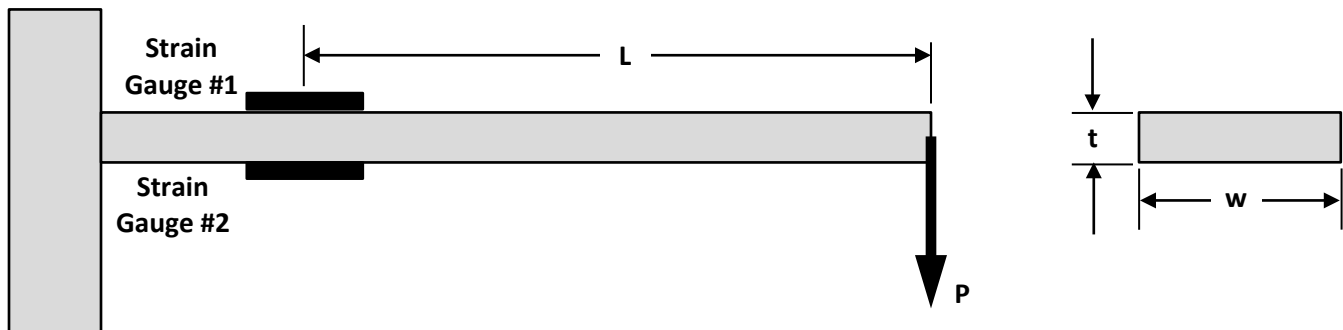


Figure 1. End loaded cantilever beam with two bonded strain gauges.

Laboratory Instructions (Parts 1-4 first week, Parts 5-6 second week)

Part 1, Build and Test a Half Arm Wheatstone Bridge Circuit

- 3) Create a Simulink program to acquire and display the output voltage from the bridge circuit. This will be most useful during circuit debugging. The primary data source, however, will be the table-top DMM.
- 4) On a breadboard, build a half arm Wheatstone bridge circuit as shown in Figure 2. Use the two strain gauges mounted to the cantilevered beam as resistors R#1 and R#2. Use matching precision resistors ($120\ \Omega$ or $350\ \Omega$) for R.

- 5) The outputs from the bridge are the inputs to the instrumentation amplifier. Use a 102 Ω resistor to set AD622 amplifier gain to $G = 500$. This will amplify the voltage to a range suitable for the table-top multi-meter. Connect the output of the instrumentation amplifier to the table-top multi-meter and the USB-DAQ.
- 6) Make sure to connect the excitation ground to the REF pin on the instrumentation amplifier (Figure 2).
- 7) Provide the amplifier and the bridge with a 10V excitation voltage (Figure 2). Use the table-top DMM and Simulink to measure the amplified output voltage, $V_o G$, from the instrumentation amplifier to check that your measurement system is working properly.

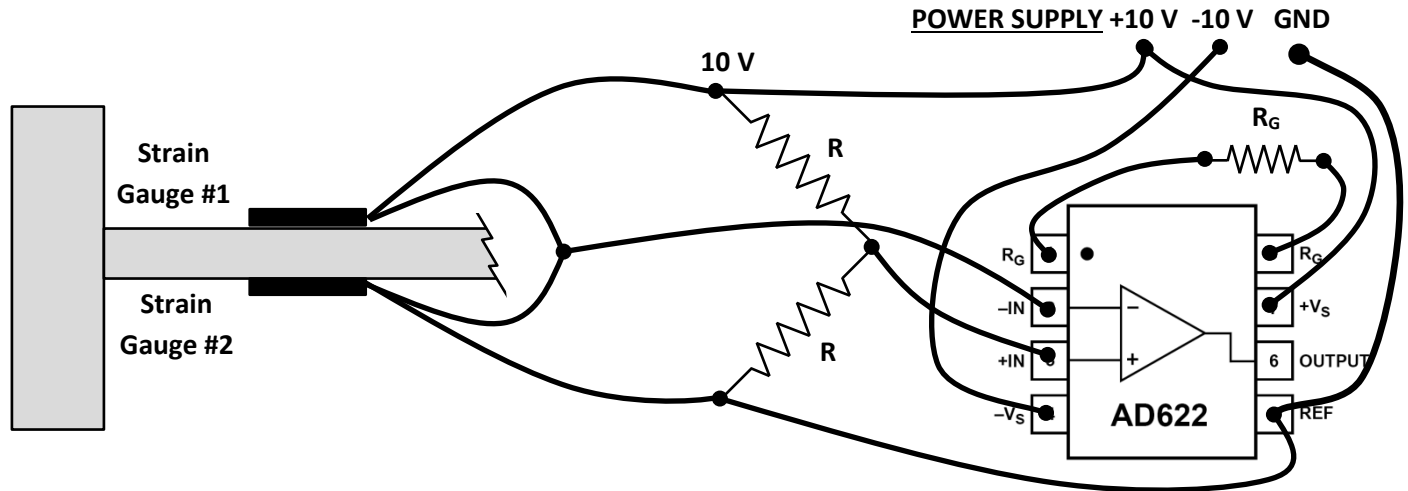


Figure 2. Two strain gauges mounted on a cantilevered beam and a half arm Wheatstone bridge circuit.

Part 2, Prepare to take Load and Strain Data for the Cantilevered Beam

- 8) Measure the length between the center of the strain gauge and point of loading, L , the width of the beam, w , and the thickness, t , and estimated all uncertainties for each. Enter these values into a Matlab script. Also, determine the local gravity by going to <https://www.ptb.de/cartoweb3/SISproject.php> and use the city search function.
- 9) Using the general equation $\sigma = My/I$, the stress at the location of the strain gauges can be found according to:

$$\sigma = \frac{My}{I} = \frac{PL(t/2)}{wt^3/12} = \frac{6PL}{wt^2} = \frac{6mgL}{wt^2}. \quad (1)$$

Add this equation for stress to your Matlab script.

- 10) The measured strain, ϵ , may be found from the $\frac{1}{2}$ Arm Wheatstone bridge circuit:

$$\frac{\delta V_o}{V_i G} = \frac{(GF)\epsilon}{2}, \quad (2)$$

where V_i is excitation voltage, $GF = 2.012$ is the gauge factor, G is the instrumentation amplifier gain, and δV_o is the change in output voltage from loaded to unloaded condition. Add this equation for strain to your Matlab script.

- 11) Create a new variable in Matlab, and open it in the array editor (name this variable `straindata`). Use this variable to record your experimental data, with loads in the 1st column (kilograms), unloaded output voltage in the 2nd column, and loaded output voltage in the 3rd column. For each load, create two separate rows in this array. Both rows will have the same load and loaded voltages, but different unloaded voltages (one from before loading, one from after removing the load). During the experiment, **repeat these data points in your logbooks to make sure no data is lost!**

Part 6, Perform an uncertainty analysis using the uncertainties in Table I.

23) How accurate is your estimate of Young's Modulus? This is the question that uncertainty analysis will attempt to quantify. For this experiment, a simplified uncertainty analysis is given by:

$$u_E = \sqrt{s_{a_1}^2 + u_E^2}, \quad (3)$$

where s_{a_1} is the standard error of the slope of the linear regression fit, and u_E is the propagation of error of the independent measurements used for the calibration data. The standard error of the slope can be found according to:

$$s_{a_1} = s_{yx} \sqrt{\frac{1}{\sum_{i=1}^N (x_i - \bar{x})^2}}, \quad (4)$$

where s_{yx} is the standard error of the the fit. To calculate the uncertainty, u_E , first estimate the uncertainty in each of the variables found in Table I. Next, u_E can be found by applying error propagation techniques to the equation for Youngs' Modulus from this experiment (found by combining equations (1) and (2)):

$$E = \frac{\sigma}{\varepsilon} = \frac{6mgL/(wt^2)}{2\delta V_o/(V_i G(GF))} = \frac{3mgLV_i G(GF)}{wt^2 \delta V_o}. \quad (5)$$

24) Derive the equation for the uncertainty in Young's modulus, E , using the general equation:

$$u_E = \sqrt{\left(\frac{\partial E}{\partial x_1} u_{x_1}\right)^2 + \left(\frac{\partial E}{\partial x_2} u_{x_2}\right)^2 + \cdots + \left(\frac{\partial E}{\partial x_n} u_{x_n}\right)^2} \quad \text{for } x = \begin{bmatrix} m & g & L & w & t & GF & G & V_i & \delta V_i \end{bmatrix} \quad (6)$$

For For m and δV_o use mean values.

25) Each term under the radical in (6) is the uncertainty propagated to the uncertainty in Young's modulus, u_E . For example, the value of the second term ($x_2 = g$) is the uncertainty in E caused by the uncertainty in g . Calculate each term in order to fill in the 3rd row of Table I. This analysis shows which variables are causing the most uncertainty in E , and thus suggests variables to measure more accurately if you wish to improve the results of the experiment.

26) In Matlab, create a 3 x 9 matrix of values from Table I. Values should be in the same order (row and column) as the table. Name the matrix `uncertaintydata`.

Post-Laboratory Assignment

First week: submit the following items on BbLearn for your post-laboratory assessment.

- 1) **Stress vs. Strain plot with linear regression.** Submit a single figure with stress (in MPa) plotted vs strain (in micro-strain). Also include your linear regression line, of which the slope is Young's modulus. Use the text command to place your experimentally determined value (in GPa) for Young's modulus on the plot.
- 2) **A `straindataETW.mat` file.** Save your `straindata` array in a `.mat` file with your initials on the end and submit it on BbLearn. Use your initials, not mine!
- 3) **Answer all questions in the post-lab assessment on BbLearn.**

Second week: submit the following items on BbLearn for your post-laboratory assessment.

- 1) A `uncertaintydataETW.mat` file. Save your `uncertaintydata` array in a `.mat` file with your initials on the end and submit it on BbLearn. Use your initials, not mine! 
- 2) **Answer all questions in the post-lab assessment on BbLearn.**

Post-Laboratory Report

This experiment will be the subject of the 2nd of 3 Formal Reports. For this second report, start with the report draft/outline posted on BbLearn.

AD622 Instrumentation Amplifier

The AD622 DIP (dual in-line package) contains several operational amplifiers arranged in an instrumentation amplifier circuit utilizing precision resistors in order to improve the accuracy of the gain and to reduce uncertainty. A connection diagram is given in below. Refer to the datasheet below for determining how to change the gain through the external resistor R_G .

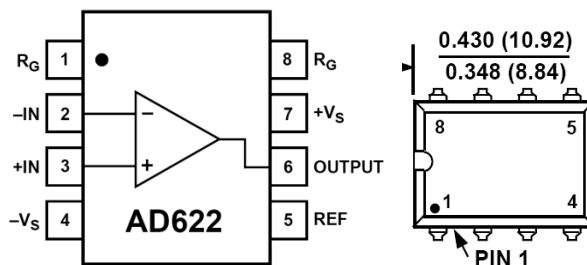


Figure 3. Connection diagram and package outline for the AD622 instrumentation amplifier.

A detailed datasheet for the AD622 can be found at:

http://www.analog.com/static/imported-files/data_sheets/AD622.pdf